EmbeddingsBench — findings

Text classification with `jinaai/jina-embeddings-v5-text-small-classification`: classifiers on frozen embeddings, TF-IDF, ensembles, LoRA fine-tuning and a German-native reference model on 18 English and German datasets.

Status: interim report, results as of 12 October 2026. Runs still in progress are listed at the end; the numbers here will be updated when they finish. All accuracies are on held-out test splits; every setting (C, k, MLP shape, ensemble weight, fine-tuning checkpoint) was chosen on the validation split only.

Key findings

1. **Logistic regression is the right default on frozen embeddings.** Across 18 datasets a tuned MLP is on average +0.2 points better (median 0.0), an RBF SVM −0.9, kNN −6.5. A 40-trial MLP architecture search most often picks a linear model.
2. **The cheapest real gain is an ensemble with TF-IDF.** Averaging LogReg probabilities with a word-n-gram TF-IDF SVM is better than LogReg on 12 of 17 datasets, +3.7 points on average, with no GPU training. It matches or beats LoRA fine-tuning on 4 of 6 datasets where both were run.
3. **Frozen embeddings struggle with many fine-grained classes.** With 342 food categories (Open Food Facts DE) LogReg reaches 44.9 % while TF-IDF reaches 57.8 % and the ensemble with character n-grams 60.6 %. For 408 patent classes (HUPD): 50.2 / 52.3 / 54.3 %. Intent datasets with 60–150 classes are the exception (87–95 %), because their classes differ in meaning, not in vocabulary.
4. **Long, technical or legal texts favour lexical features.** On Swiss court decisions, arXiv and LEDGAR, TF-IDF alone beats LogReg on frozen embeddings by 3–7 points and is within 1.5 points of the best method.
5. **LoRA fine-tuning helps most on topical datasets with mid-length texts** (+3.7 to +7.0 points over LogReg on Yahoo! Answers, 20 Newsgroups, 10kGNAD, arXiv), little on short texts with semantic classes (MASSIVE +0.0, Amazon stars +1.4). On 10kGNAD the German-native `gbert-large` (fully fine-tuned) is ahead: 91.3 % vs 88.6 % (89.6 % with head + tail truncation).
6. **Zero-shot (label-name similarity) only works when label names are descriptive**: 91 % on emotions, 82 % on banking intents, under 10 % on legal areas, patents and product categories.
7. **For imbalanced data, class weighting improves the rare classes at no cost in accuracy** (LEDGAR: tail recall 67.0 → 69.8 %, balanced accuracy 71.5 → 72.7 %); logit adjustment selected $\tau = 0$, i.e. no change.
8. **2,048-token input only matters when most texts are long** (Swiss decisions +2.2, 10kGNAD +0.9, arXiv +0.5 points vs 512 tokens). **Matryoshka truncation to 512 dimensions costs 1.1 points on average** (at most 3.1), 256 dimensions 2.9 points.

Setup

- **Encoder:** `jinaai/jina-embeddings-v5-text-small-classification` (revision `4447914`), last-token pooling, L2-normalised 1,024-d embeddings, prompt `Document:`, up to 2,048 tokens.
- **Classifiers on frozen embeddings:** zero-shot (cosine similarity to label-name embeddings), kNN (cosine, k tuned), logistic regression (C tuned), RBF SVM, MLP (1–2 GELU layers) and a 40-trial MLP architecture search.
- **Text baselines:** TF-IDF (word 1–2-grams, sublinear tf) with a linear SVM; character 2–5-gram TF-IDF.
- **Ensembles:** weighted average of LogReg probabilities and softmaxed TF-IDF scores; weight and temperature chosen on validation.

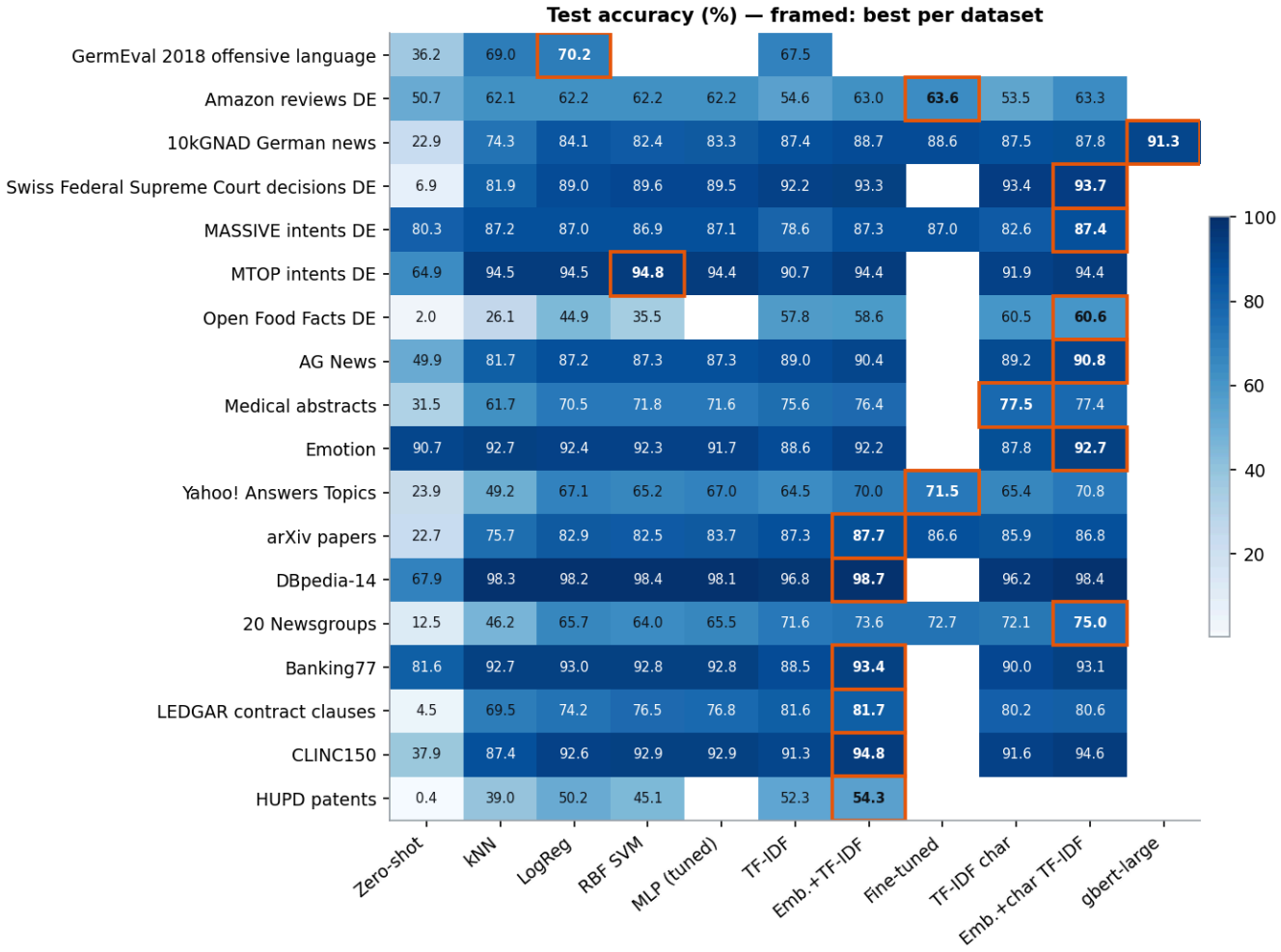
- **Fine-tuning:** LoRA rank 16 on all attention and MLP projections, linear head initialised from the frozen LogReg, 3 epochs, 512 tokens (variants: 1,024 tokens, head + tail truncation, class-weighted and logit-adjusted loss). `deepset/gbert-large` fully fine-tuned as a German reference.
- **Metrics:** accuracy, balanced accuracy, macro-F1, MCC, Cohen's κ, top-3 accuracy and tail recall (recall on the third of classes with the fewest training examples).
- **Hardware:** one RTX 3060 (12 GB), shared with other services.

Datasets

Dataset	Lang	Domain	Classes	Train / val / test	Avg tokens	Train imbalance
GermEval 2018 offensive language	DE	tweets (offensive language)	4	4,258 / 751 / 3,398	45	47:1
Amazon reviews DE (stars)	DE	product reviews (sentiment)	5	10,000 / 5,000 / 4,999	65	1:1
10kGNAD German news	DE	news	9	7,860 / 1,384 / 1,027	732	3:1
Swiss Federal Supreme Court decisions DE (legal area)	DE	court decisions	47	25,737 / 3,002 / 3,002	3413	19:1
MASSIVE intents DE	DE	assistant intents	60	11,514 / 1,943 / 2,845	12	202:1
MTOP intents DE	DE	assistant intents (long tail)	103	11,072 / 1,956 / 1,756	14	1175:1
Open Food Facts DE (product categories)	DE	food products	342	69,366 / 8,790 / 8,790	18	12:1
AG News	EN	news	4	10,000 / 2,000 / 7,599	54	1:1
Medical abstracts	EN	medical	5	9,817 / 1,221 / 1,878	274	3:1
Emotion (tweets)	EN	emotion (tweets)	6	16,000 / 1,995 / 1,986	22	9:1
Yahoo! Answers Topics	EN	community Q&A	10	10,000 / 2,000 / 5,000	127	1:1
arXiv papers (11 categories)	EN	scientific papers	11	8,800 / 2,446 / 2,441	4300	1:1
DBpedia-14	EN	encyclopedia	14	7,000 / 1,400 / 2,800	79	1:1
20 Newsgroups	EN	forum posts	20	9,361 / 1,644 / 7,299	275	2:1
Banking77	EN	banking intents	77	8,497 / 1,496 / 3,069	15	5:1
LEDGAR contract clauses (LexGLUE)	EN	legal contracts	100	19,092 / 9,914 / 9,855	144	9:1
CLINC150 (in-scope)	EN	assistant intents	150	15,000 / 2,997 / 4,498	11	1:1
HUPD patents (IPC subclass)	EN	patents (603-way long tail)	408	98,195 / 18,934 / 18,901	151	50:1

Splits are fixed and stratified where the source has none; large sources are capped per class (see `bench/data.py`). Train imbalance is the ratio of the largest to the smallest class in the training split.

Overview

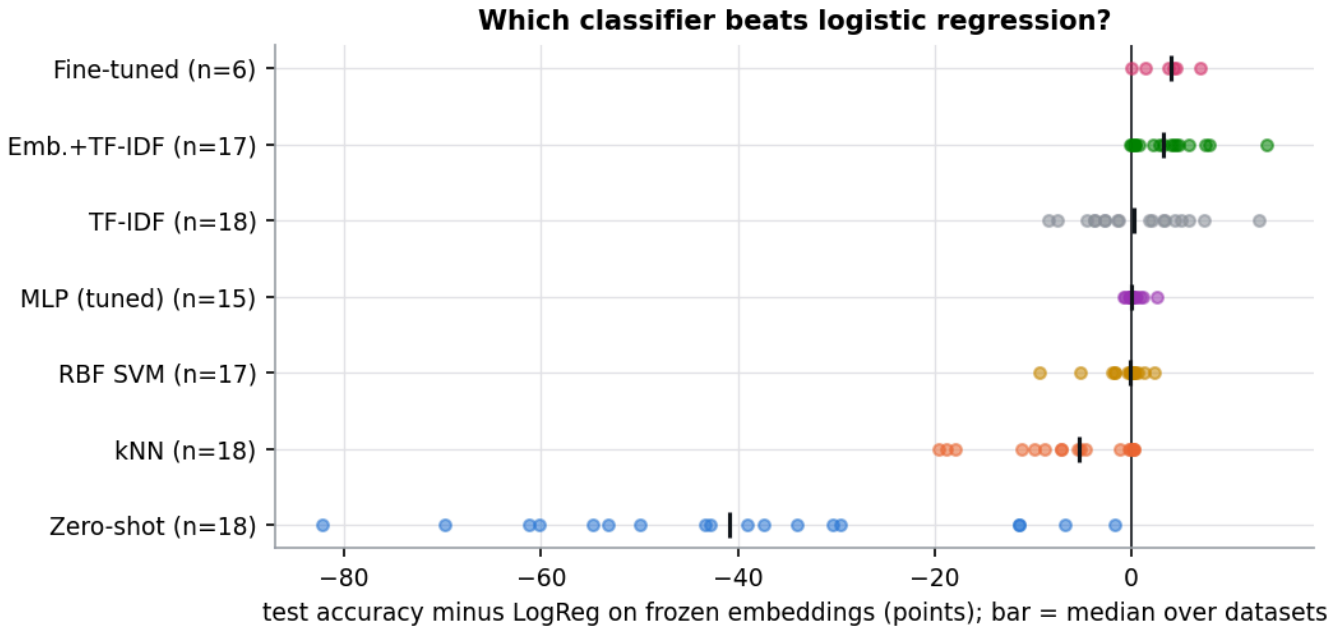


Dataset	Classes	Zero-shot	kNN	LogReg	MLP	TF-IDF	Emb.+TF-IDF	Best (method)	Macro-F1 of best
GermEval 2018 offensive language	4	36.2	69.0	70.2	70.3	67.5	–	**70.3** (MLP)	37.1
Amazon reviews DE	5	50.7	62.1	62.2	62.2	54.6	63.0	**63.6** (Fine-tuned jina)	63.1
10kGNAD German news	9	22.9	74.3	84.1	83.3	87.4	88.7	**91.3** (gbert-large)	91.2
Swiss Federal Supreme Court decisions DE	47	6.9	81.9	89.0	89.5	92.2	93.3	**93.7** (Emb.+char TF-IDF)	89.4
MASSIVE intents DE	60	80.3	87.2	87.0	87.1	78.6	87.3	**87.4** (Emb.+char TF-IDF)	83.5
MTOP intents DE	103	64.9	94.5	94.5	94.4	90.7	94.4	**94.8** (RBF SVM)	83.2
Open Food Facts DE	342	2.0	26.1	44.9	45.4	57.8	58.6	**60.6** (Emb.+char TF-IDF)	56.9
AG News	4	49.9	81.7	87.2	87.3	89.0	90.4	**90.8** (Emb.+char TF-IDF)	90.7
Medical abstracts	5	31.5	61.7	70.5	71.6	75.6	76.4	**77.5** (TF-IDF char)	77.2
Emotion	6	90.7	92.7	92.4	91.7	88.6	92.2	**92.7** (Emb.+char TF-IDF)	88.1
Yahoo! Answers Topics	10	23.9	49.2	67.1	67.0	64.5	70.0	**71.5** (Fine-tuned jina)	71.2

Dataset	Classes	Zero-shot	kNN	LogReg	MLP	TF-IDF	Emb.+TF-IDF	Best (method)	Macro-F1 of best
arXiv papers	11	22.7	75.7	82.9	83.7	87.3	87.7	**87.7** (Emb.+TF-IDF)	87.0
DBpedia-14	14	67.9	98.3	98.2	98.1	96.8	98.7	**98.7** (Emb.+TF-IDF)	98.7
20 Newsgroups	20	12.5	46.2	65.7	65.5	71.6	73.6	**75.0** (Emb.+char TF-IDF)	73.4
Banking77	77	81.6	92.7	93.0	92.8	88.5	93.4	**93.4** (Emb.+TF-IDF)	93.3
LEDGAR contract clauses	100	4.5	69.5	74.2	76.8	81.6	81.7	**81.7** (Emb.+TF-IDF)	77.6
CLINC150	150	37.9	87.4	92.6	92.9	91.3	94.8	**94.8** (Emb.+TF-IDF)	94.8
HUPD patents	408	0.4	39.0	50.2	50.4	52.3	54.3	**54.3** (Emb.+TF-IDF)	38.7

The best method per dataset is picked by test accuracy for this overview; each method's own settings were chosen on validation. On GermEval 2018 and the long-tail sets (MTOP, HUPD), macro-F1 is far below accuracy: the majority classes dominate accuracy.

Classifiers on top of the embeddings

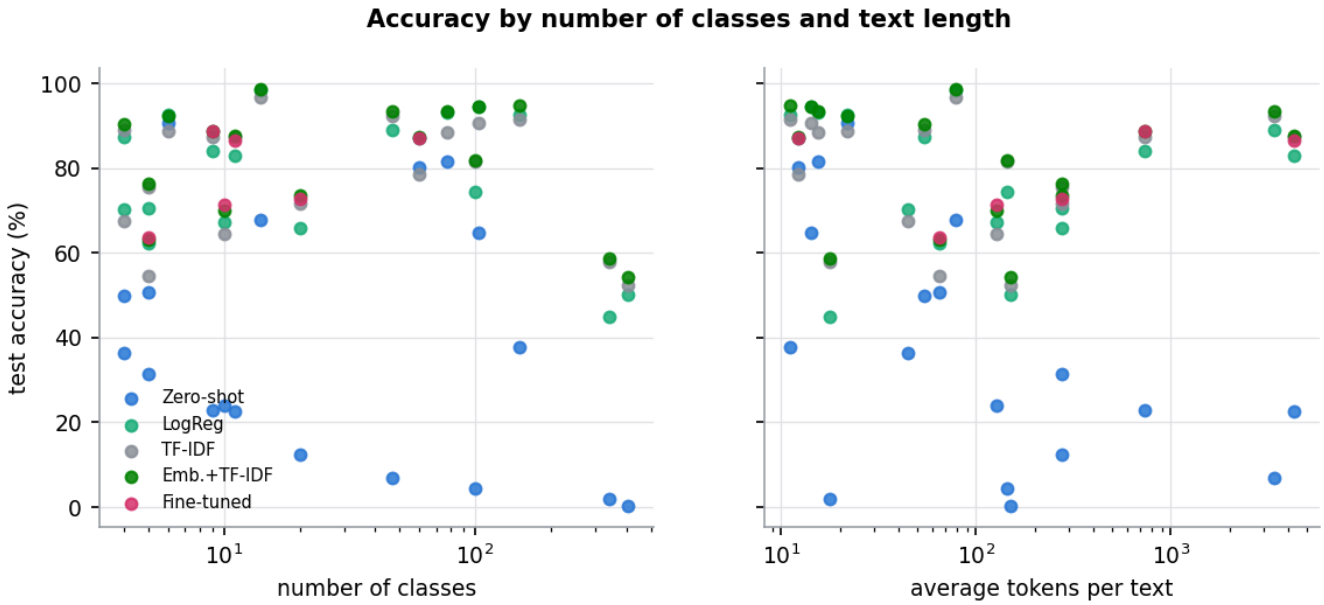


Test accuracy difference to LogReg on frozen embeddings, in points:

Classifier	Mean Δ (pts)	Median Δ (pts)	Better by > 0.5 pts	Datasets
Emb.+TF-IDF	+3.7	+3.2	12	17
Fine-tuned	+3.5	+4.0	5	6
TF-IDF	+0.5	+0.3	9	18
MLP (tuned)	+0.2	+0.0	3	15
RBF SVM	-0.9	-0.1	3	17
kNN	-6.5	-5.3	0	18
Zero-shot	-39.9	-40.9	0	18

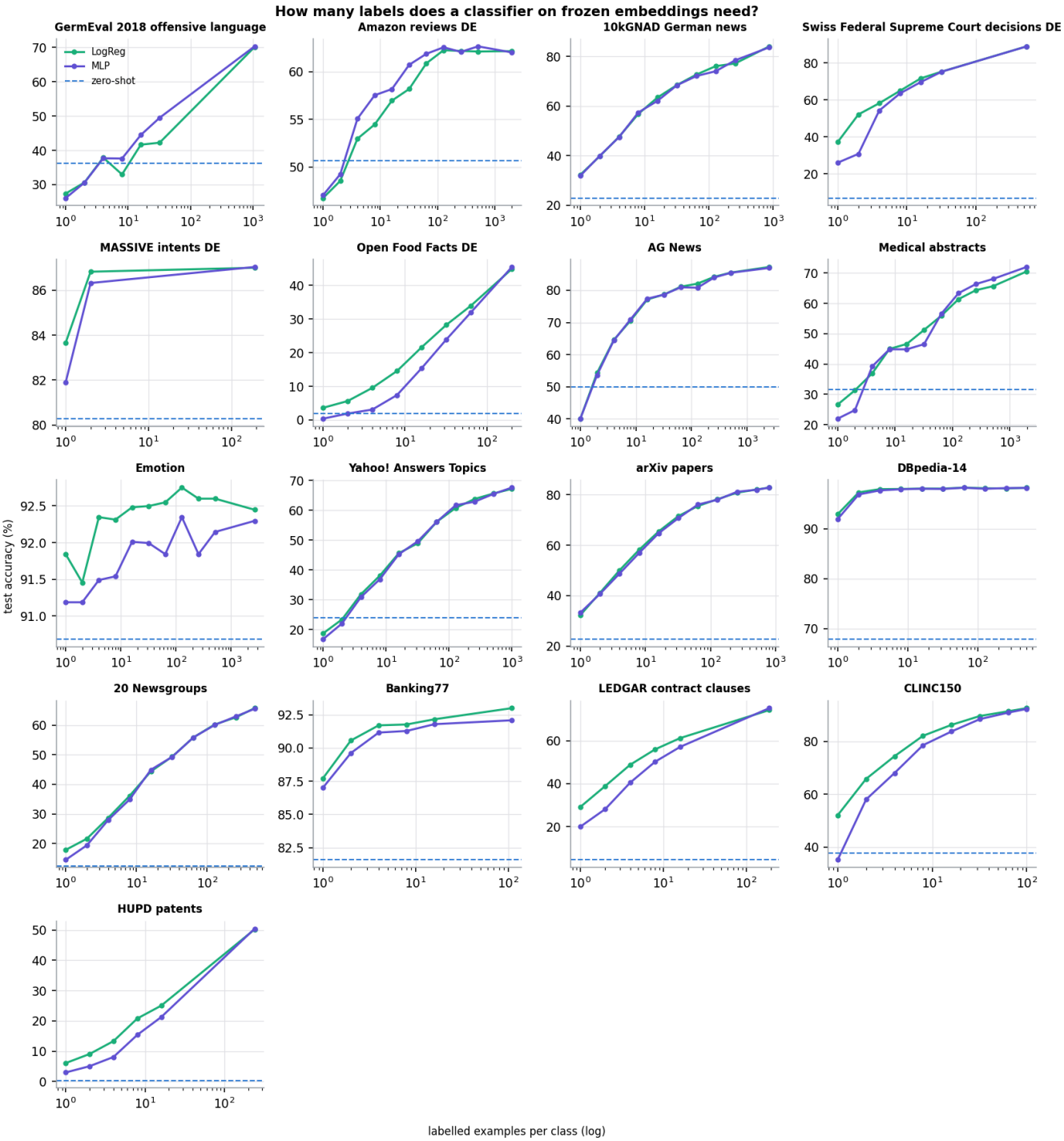
- **LogReg vs MLP:** the embeddings are already linearly separable where they separate at all. MLPs win by 1–2 points only on a few datasets with larger training sets (LEDGAR +2.6 tuned, Medical abstracts +1.1) and lose on small or many-class sets; the search mostly chose depth 0.
- **RBF SVM:** never clearly better than LogReg, slower to train (a 30k subsample was needed above 30k training texts) and on Open Food Facts 9 points worse.
- **kNN:** competitive on intents and emotions (within 0.5 points of LogReg on Banking77, MTOP, MASSIVE, Emotion), weak on topical and many-class data (−18 points on Yahoo!, −19 on Open Food Facts).
- **Zero-shot:** useful as a no-label starting point only where label names describe the class well.

Number of classes and text length



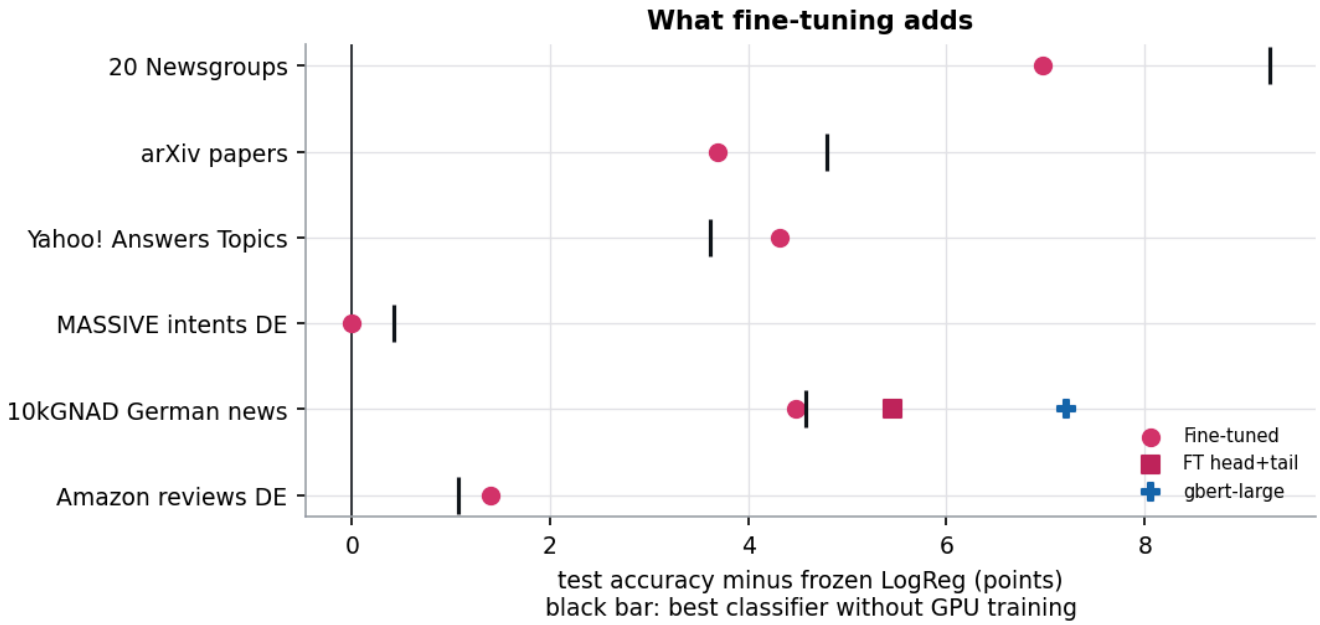
Accuracy on frozen embeddings drops sharply above about 150 classes, while TF-IDF degrades less: a single 1,024-d embedding appears to blur fine distinctions between neighbouring product or patent categories that specific words and word parts still separate. Text length alone is not the driver: short intent texts are classified well and short product names badly.

How many labels are needed



With 8–16 examples per class, LogReg on frozen embeddings already reaches most of its full-data accuracy on intent, emotion and encyclopedia data (Banking77 ~92 % at 16 per class, DBpedia ~98 % at 4). Topical and many-class datasets keep improving up to the full training set; there, more labels pay off more than a different classifier. LogReg is mostly ahead of the MLP at small sample sizes.

Fine-tuning



Dataset	Frozen LogReg	Best without GPU training	Fine-tuned jina (LoRA)	Variant	gbert-large	Training time
Yahoo! Answers	67.1	70.8	**71.5**	–	–	39 min
20 Newsgroups	65.7	**75.0**	72.7	–	–	46 min
arXiv (11)	82.9	**87.7**	86.6	–	–	150 min
10kGNAD	84.1	88.7	88.6	89.6 (head + tail)	**91.3**	76 / 95 / 29 min
Amazon reviews DE	62.2	63.3	**63.6**	–	running	37 min
MASSIVE DE	87.0	**87.4**	87.0	–	queued	26 min

- Fine-tuning pays off where classes are topical and the frozen embedding is not tuned to the distinction (Yahoo!, 20NG, news). On MASSIVE the best checkpoint was step 0, i.e. the untouched encoder.
- Where lexical cues matter (20NG, arXiv), the embedding + TF-IDF ensemble beats the fine-tuned encoder without any GPU training; the fine-tuned + TF-IDF ensemble is being computed.
- Keeping the end of long German news articles (head + tail: first 384 + last 128 tokens) adds 1.0 point over the first 512 tokens.
- `gbert-large` (335M parameters, all weights trained, 8-bit AdamW, about 4 GB of GPU memory) beat LoRA on jina by 2.7 points on 10kGNAD in less than half the training time; Amazon DE and MASSIVE DE follow.

Class imbalance

Rare classes: recall on the third of classes with the fewest training examples

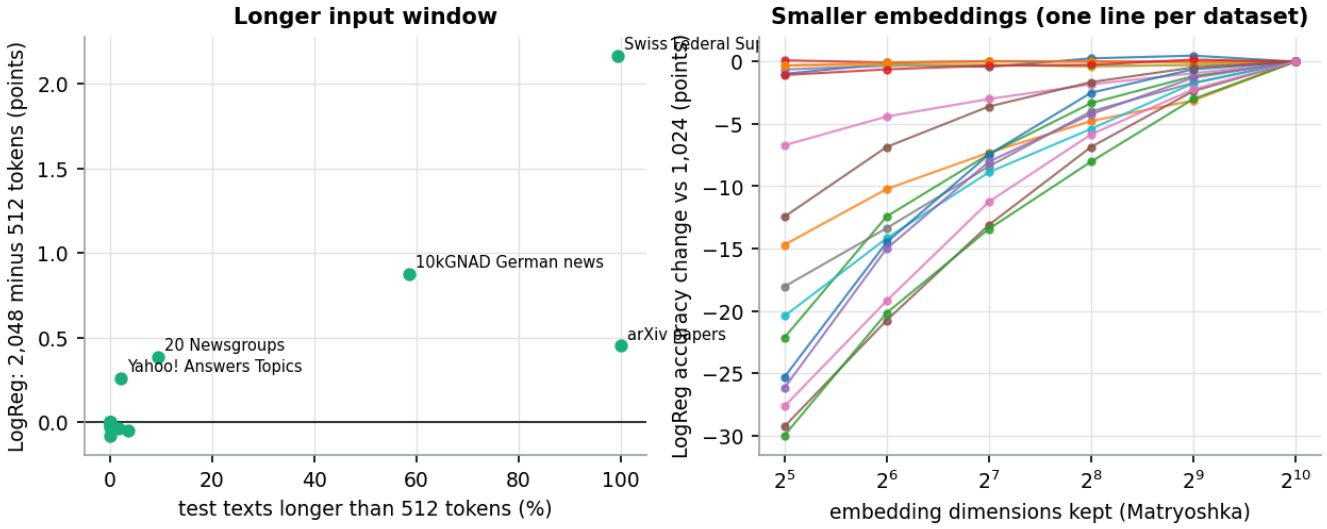


LEDGAR contract clauses (LexGLUE), %

Classifier	Accuracy	Balanced acc.	Macro-F1	Tail recall
LogReg	74.2	71.5	68.3	67.0
LogReg weighted	74.1	72.7	68.8	69.8
Logit-adjusted	74.2	71.5	68.3	67.0
MLP (tuned)	76.8	72.6	69.7	68.1
MLP weighted	76.4	72.7	69.3	69.2
MLP balanced	74.5	71.8	68.4	68.8
TF-IDF	81.6	80.0	77.5	77.4
Emb.+TF-IDF	81.7	80.1	77.6	77.4

On LEDGAR (100 clause types, training set capped at 200 per class, 9:1 imbalance), class-weighted LogReg raises recall on the rarest third of classes by 2.8 points and balanced accuracy by 1.2 points without losing accuracy. Logit adjustment chose $\tau = 0$ on validation. The lexical model is the strongest on rare classes here (tail recall 77.4 %). Imbalance schemes on MTOP (1175:1), Swiss decisions, Open Food Facts and HUPD, including class-weighted and logit-adjusted fine-tuning, are running.

Input length and embedding size



The longer window helps only in proportion to how many texts exceed 512 tokens. Matryoshka truncation: 512 dimensions −1.1 points on average (worst −3.1), 256 −2.9, 128 −5.5. The loss below 256 dimensions varies widely between datasets (see the figure).

Cost

Embedding runs at about 12,700 tokens/s and LoRA fine-tuning at about 1,900 tokens/s on the shared RTX 3060; both scale with tokens, not with the number of classes. LogReg, kNN and TF-IDF train in seconds to minutes on CPU, except TF-IDF with character n-grams on large many-class sets (HUPD: more than 11 GB of RAM, skipped).

Recommendations

Default pipeline. Embed once, train LogReg (tune C on a validation split), and add a TF-IDF SVM in a probability ensemble. That is the best or within a point of the best on most datasets, costs one embedding pass, and retrains in minutes when labels change.

Larger datasets (≫ 10k training texts). - Embed once and keep the embeddings: classifier experiments then cost CPU minutes, not GPU hours. - Fit LogReg with a fast solver (SAGA or a GPU softmax layer) and a coarse C grid; subsample for SVMs and for hyper-parameter search. - An MLP becomes worth trying above roughly 20k training texts (LEDGAR +2.6 points), still with a 1-layer default. - Fine-tune only on a capped subset first (e.g. 1,000 per class): the learning curves show diminishing returns, and fine-tuning time grows linearly with tokens. - Use 512-d Matryoshka embeddings when storage or kNN search matters; budget about 1 point of accuracy.

Many classes (> 100–150). - Do not rely on frozen embeddings alone: always include TF-IDF (word and character n-grams) and the ensemble; on 342 and 408 classes it was 4–16 points ahead of LogReg. - Report balanced accuracy, macro-F1 and top-3 alongside accuracy; macro-F1 on HUPD is 16 points below accuracy. - Make sure each class has enough examples (≥ 20, ideally ≥ 100); merge or drop catch-all and near-duplicate categories (as done for Open Food Facts). - Consider a hierarchy: predict the parent category first (e.g. the IPC class), then the subclass among its children. - Use class weighting, and check tail recall. - Zero-shot and kNN are not viable at this scale.

Short texts with semantic classes (intents, emotions). Frozen embeddings + LogReg are enough (87–95 %); kNN is a good alternative when classes change often, because no retraining is needed. Fine-

tuning brings little.

Long documents. Use the 2,048-token window (or head + tail when fine-tuning) and combine with TF-IDF; legal and scientific texts carry much of their signal in vocabulary.

German. jina v5 handles German well on frozen embeddings, but for a fine-tuned German classifier a German-native encoder (`gbert-large`) was stronger in the first comparison.

Imbalanced data. Start with class-weighted LogReg (free), measure tail recall, and only then try logit adjustment or weighted fine-tuning.

Still running

- gbert-large on Amazon reviews DE and MASSIVE DE.
- MLP search, imbalance schemes, GermEval extras on the new datasets.
- 10kGNAD fine-tuning with 1,024 tokens; fine-tuned + TF-IDF ensembles.
- Fine-tuning (plain, class-weighted, logit-adjusted) on MTOP and Swiss decisions.

Reproduce

Code, data preparation and all raw results are in this repository: `results/results.json` (every number above, incl. per-setting validation scores), `bench/` (pipeline), `bench/report.py` (these figures and tables). See the README for setup and the GPU server configuration.